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BFLPE, Social Comparison Costs, and Academic Self-Concept in Gifted and Selective Settings

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Background: BFLPE and social comparison in academic self-concept

The big-fish-little-pond effect says that equally able students often feel less academically capable when they are placed in higher-achieving schools or classes. The core reason is social comparison: students judge themselves partly against the achievement level of the peers around them, not just against their own actual performance. (18 sources)

The big-fish-little-pond effect (BFLPE) is a notable finding in research on academic self-concept: students in high-achieving schools or classes tend to report lower academic self-concept than equally able students in lower-achieving settings. Put simply, it is often better for self-concept to be a “big fish” in a lower-achieving pond than a “small fish” in a higher-achieving one. This pattern has been described across schools, classrooms, and selective settings, and it appears robust across many countries and study designs. (Jackman et al., 2011) (Becker et al., 2016) (Wang, 2020) (Papadopoulos, 2021) (Zhang, 2021) (Basarkod et al., 2020) (Mai et al., 2022) (Lesick et al., 2023) (Hasenbein et al., 2023) (Becker et al., 2024) (Shahbazian, 2026) (Rathmann et al., 2018) (Marsh et al., 2008)

The basic logic is a social frame-of-reference account. A student’s own achievement usually raises self-concept at the individual level, but the average achievement of classmates or schoolmates can lower it at the context level because students compare their performance with the local norm. In a lower-achieving group, the same student may feel above average and more competent; in a higher-achieving group, that student may feel below average and less competent, even when actual ability is unchanged. (Arens et al., 2015) (Goetz et al., 2008) (Hoffler et al., 2017) (Hoffler et al., 2016) (Becker et al., 2024)

This matters because academic self-concept is not a trivial feeling; it is linked to later outcomes such as achievement, course choices, educational attainment, and mental development. For that reason, the BFLPE is not just a theory about perception. It identifies a real comparison cost that can come with high-ability contexts, even when those contexts may offer other academic benefits. (Jackman et al., 2011) (Papadopoulos, 2021) (Fang et al., 2018) (Marsh et al., 2008)

Evidence

(Jackman et al., 2011)

Big Fish in a Big Pond: a study of academic self concept in first year medical students

"Big-fish-little-pond effect (BFLPE) research has demonstrated that students in high-ability environments have lower academic self-concepts than equally able students in low-ability settings. Research has shown low academic self-concepts to be associated with negative educational outcomes. Social comparison processes have been implicated as fundamental to the BFLPE."

(Becker et al., 2016)

Context-related changes in academic self concept development: On the long-term persistence of big-fish-little-pond effects

"Research on the so-called big-fish-little-pond effect (BFLPE; (Marsh, 1984)(Marsh, 1987) provided evidence that individuals who are exposed to a high-achieving reference group perceive themselves as academically less able and less competent than do equally able individuals who are exposed to a low-achieving reference group."

(Wang, 2020)

When Large-Scale Assessments Meet Data Science: The Big-Fish-Little-Pond Effect in Fourth- and Eighth-Grade Mathematics Across Nations

"The BFLPE refers to the phenomenon that students in higher-achieving contexts tend to have lower self-concept than similarly able students in lower-achieving contexts due to social comparison."

(Papadopoulos, 2021)

Examining the Relationships among Cognitive Ability, Domain-Specific Self-Concept, and Behavioral Self-Esteem of Gifted Children Aged 5–6 Years: A Cross-Sectional Study

"This phenomenon is known as the "Big-Fish-Little-Pond Effect" (BFLPE); it states that able students in high-ability environments have lower academic self-concept than those students with equal abilities in lower-ability settings. That is, being a big fish in a small pond benefits one's academic self-concept. Research has demonstrated that the BFLPE is likely the result of students comparing their competence to the average ability of their peers (Fang et al., 2018)."

(Zhang, 2021)

Big fish in a big pond: Peer effects on university students' academic self-concept

"According to the big-fish-little-pond effect (BFLPE), students enrolled in the high-peerachievement schools (classes) had lower ASC compared with equally capable students engaged in the low-peer-achievement learning environment (Marsh, 1984)(Marsh , 1987)). Several studies have testified the existence of the BFLPE, and most researchers believed that personal achievements had positive effects on ASC while peer achievements had negative impact on ASC (Lüdtke et al., 2005)."

(Basarkod et al., 2020)

The Big-Fish-Little-Pond Effect for Reading Self-Beliefs: A Cross-National Exploration with PISA 2018

"Past research shows the Big-Fish-Little-Pond Effect (BFLPE; negative effect of school-average achievement on student-level academic self-concept) to generalize across countries"

".Conclusions Our findings extend support for the generalizability of the BFLPE to reading self-concept and highlight the role of social comparison processes underlying this effect."

(Mai et al., 2022)

Achievement Emotions in Selective Schools: Reexamining the Happy-Fish-Little-Pond Effect in an Extreme Case from the Chinese Collectivist Context

"The BFLPE describes the well-known phenomenon that students in selective schools/classes (big ponds) tend to have lower academic self-concepts than those who are comparably competent but attend regular schools/classes (little ponds), and this effect was shown to be stable and intercultural using considerable evidence (Marsh et al., 2003) and meta-analyses (Fang et al., 2018). The mechanism at the heart of these two effects lies in social comparison (Marsh, 1987). In addition to using objective, physical, and non-social criteria to evaluate their performance, individuals evaluate themselves by comparing themselves with others (Festinger, 1954). To some extent, feedback from the outside world is more important than objective self-evaluation standards (Klein, 1997). At school, peers who pursue similar or common goals are the main objects of comparison (Huang et al., 2019). According to the frame-of-reference effect, students' academic self-concepts and achievement emotions depend not only on their absolute academic level but also on their level relative to other students in the same school/class (Marsh, 1986). In selective schools/classes, being ranked low results in low academic self-concepts and more negative achievement emotions in students."

(Lesick et al., 2023)

Race-ethnicity and the big-fish-little-pond effect in the United States

"Research on the big-fish-little-pond effect (BFLPE) demonstrates that students with a high rank in a low rank school have more

favorable self-evaluations than students with a low rank in a high rank school (Marsh et al., 1984)(Marsh et al., 2019)."

(Hasenbein et al., 2023)

An experimental test of the Big-Fish-Little-Pond Effect using an immersive virtual reality classroom

"The BFLPE suggests that unless students are "the big fish in the pond" (i.e., above the average performance level of their reference group), they are constantly exposed to comparisons with more able peers and consequently tend to evaluate themselves as less competent than they would in a lower achieving class (Marsh et al., 2003)(Marsh et al., 2021)(Nagengast et al., 2012)(Seaton et al., 2009)Seaton et al., , 2010)). In other words, the BFLPE suggests that two equally able students will evaluate themselves differently depending on the performance level of their classmates, whom they use as a reference for their self-evaluation."

(Becker et al., 2024)

Living in the Big Pond: Adding the Neighborhood as a Frame of Reference for Academic Self-Concept Formation

"students' academic self-concept is negatively predicted by the average achievement of educational environments, that is, schools or classrooms when controlling for individual achievement (for an overview, see (Marsh et al., 2015). In other words, equally able students have lower academic self-concepts in high-achieving environments. This finding is called the bigfish-little-pond effect (BFLPE; (Marsh, 1987). The BFLPE emerges as a consequence of social comparison processes in which students evaluate their academic capabilities by comparing their achievement with that of other students in their educational environment (e.g., (Huguet et al., 2009)Marsh et al., 2014). The psychological mechanism causing the BFLPE is a contrast effect, implying that a student of given achievement forms contrasts with a generalized other-that is, the classroom or school environment-to fulfill comparison needs (Marsh et al., 2000)(Marsh et al., 2008). Thus, if a student assumes that this generalized other's achievement is superior (inferior), it harms (fosters) the student's self-concept."

(Shahbazian, 2026)

Ranked and at risk: Relative academic position in school and receiving social assistance in adulthood.

"This effect describes how pupils with similar academic abilities can develop very different self-concepts depending on the achievement level of their peers (Marsh, 1987). Pupils surrounded by high-achieving classmates are more likely to develop a negative view of their own academic ability, whereas pupils who perform well in a lower-achieving context often feel more competent and confident."

(Rathmann et al., 2018)

Class-Level School Performance and Life Satisfaction: Differential Sensitivity for Low- and High-Performing School-Aged Children

"Most prominent in relation to social comparisons between students in educational settings, studies have highlighted that equally able students have lower academic self-concepts when attending schools or classes where the average ability level of classmates is higher, and higher academic self-concepts when attending schools or classes where the school-or class-average ability is lower (Belfi et al., 2012)[8](Trautwein et al., 2006)(Trautwein et al., 2009)(Marsh, 2016)(Marsh et al., 2008). This association is well-known as the so-called "contrast-" or "big-fish-little-pond"-effect (BFLPE). According to the BFLPE-hypothesis, it is argued that it is better for school-aged children's general or academic self-concept to be a "big fish" in a "little pond" (i.e., in reference groups or classes with on average lower performance levels) than to be a "little fish" in a "big pond" (i.e., in reference groups or classes with on average higher performance levels). In other words, a student will have a lower (academic) self-concept ("little fish") in a learning group with higher school performance ("big pond") and will have a higher (general or academic) self-concept ("big fish") in a class with lower school performance ("little pond") because this student compares and contrasts his or her own school performance with that of classmates. Empirical support for this contrast-or BFLP-effect comes from manifold national and cross-national studies, for different educational settings (Seaton et al., 2009)(Marsh, 2016)21,(Salchegger, 2016) and mainly in relation to young people's general or academic self-concept as well as self-esteem (Trautwein et al., 2006). Findings on the BFLPE are remarkably robust, generalizing over a wide variety of different individual student and contextual level characteristics, settings, countries, long-term follow-ups, and research designs [17,20]23]."

(Marsh et al., 2008)

The Big-fish–little-pond-effect Stands Up to Critical Scrutiny: Implications for Theory, Methodology, and Future Research

"The big-fish–little-pond effect (BFLPE) predicts that equally able students have lower academic self-concepts (ASCs) when attending schools where the average ability levels of classmates is high, and higher ASCs when attending schools where the school-average ability is low"

".Because of the importance of ASC in predicting future achievement, coursework selection, and educational attainment, the results have important implications for the way in which schools are organized (e.g., tracking, ability grouping, academically selective schools, and gifted education programs)"

".The flaw in this argument, as emphasized by (Hattie, 2002), is confounding the negative effects of ability grouping per se (the focus of the BFLPE) with the many other components that are likely to be incorporated into gifted education programs (e.g., different curriculums; more dedicated, highly trained teachers; better resources; enrichment experiences) that might be expected to

have positive effects on ASC."

(Arens et al., 2015)

How an early transition to high-ability secondary schools affects students' academic self-concept: Contrast effects, assimilation effects, and differential stability

"Contrast effects on students' academic self-concept have been investigated primarily in the context of research on the big-fish-little-pond effect (BFLPE; e.g., (Marsh, 1987). In this case, social comparison processes are assumed to yield differential effects on the individual (student) level and the group or context (class-average or school-average) level. Given that students compare their own achievement in one school subject with the achievement of other students in the same subject, high achievement leads to high levels of academic self-concept on the individual level. In addition, students compare their own achievement with the average achievement of the group to which they belong (within-group comparison; (Liu et al., 2005), leading to a negative effect of achievement on academic self-concept on the context level. Equally able students display low levels of academic self-concept in highability contexts, but high levels of academic self-concept in low-ability contexts. A student who is consistently confronted with better achieving students (i.e., with high class-average or school-average levels of achievement) might develop a poor academic self-concept, as this student always perceives his or her own accomplishments to be inferior to those of the other students. Conversely, the same student might perceive his or her accomplishments as above average in low-ability contexts, strengthening his or her academic self-concept."

(Goetz et al., 2008)

Big fish in big ponds: A multilevel analysis of test anxiety and achievement in special gifted classes

"The BFLPE hypothesizes that it is better for academic self-concept, and, consequently, for test anxiety to be a big fish in a little pond (i.e., good student in a peer reference group of average ability), than to be a small fish in a big pond (i.e., good student in a peer reference group of high ability). (Marsh et al., 1984) proposed a social frame of reference model to explain the BFLPE. According to this model, self-perceptions in educational settings are largely shaped by the process of social comparison. With increasing ability level of a peer reference group (e.g., class), students often compare themselves with high ability peers and are compared by their teachers with more intellectually able peers, which, in turn, affects the feedback (e.g., grades) students receive."

(Hoffler et al., 2017)

Science vs. Sports: Motivation and Self-concepts of Participants in Different School Competitions

"For instance, a well-established effect of social references is the big-fish-little-pond (BFLP) effect (Marsh, 1987): The mean achievement level of, for example, fellow students (the Bexternal frame of reference^; cf. internal/external frame of reference model, I/E model; (Marsh, 1986) can have a significant effect on the individual self-concept. Equally able students may develop different academic self-concepts depending on the academic abilities of their fellow students (the external frame of reference). If Bthe pond is small(i.e. the average ability level of the school's students is rather low), then an averageability student will use this reference for social comparison with the likely result that (s)he will develop a high academic self-concept (Marsh, 1987). If the same student was in a class with more able students than herself/himself, a significantly lower selfconcept would likely be the result (Marsh et al., 2008)."

(Hoffler et al., 2016)

Science vs. Sports: Motivation and Self-concepts of Participants in Different School Competitions

"One's self-evaluation of competences often refers to specific reference standards and comparisons (Heckhausen & Leppmann, 1991). These comparisons are crucial for the academic self-concept (Marsh et al., 2008). For instance, a well-established effect of social references is the big-fish-little-pond (BFLP) effect (Marsh, 1987): The mean achievement level of, for example, fellow students (the Bexternal frame of reference^; cf. internal/external frame of reference model, I/E model; (Marsh, 1986) can have a significant effect on the individual self-concept. Equally able students may develop different academic self-concepts depending on the academic abilities of their fellow students (the external frame of reference). If Bthe pond is small(i.e. the average ability level of the school's students is rather low), then an averageability student will use this reference for social comparison with the likely result that (s)he will develop a high academic self-concept (Marsh, 1987). If the same student was in a class with more able students than herself/himself, a significantly lower selfconcept would likely be the result (Marsh et al., 2008)."

(Fang et al., 2018)

The Big-Fish-Little-Pond Effect on Academic Self-Concept: A Meta-Analysis

"The Big Fish Little Pond Effect (BFLPE) is one of the most influential theories about student ASC forming process, which was proposed by Marsh (1984) to describe the phenomenon that students in selective schools always have lower ASC compared to those with comparable ability but attend regular schools, which means that being a big fish in a small pond does good to one's ASC. Considerable evidence substantiated that the BFLPE is thought to be the outcome of individuals comparing their ability with the average ability of their group (Marsh, 1987;(Plieninger et al., 2015)"

".Students compare their own achievement with that of their class-or schoolmates, which leads them to feel more negative about

their own competencies in high-achieving atmosphere than in low-achieving atmosphere."

Evidence for comparison costs in high-ability, selective, and grouped settings

The evidence is consistent: when students move into higher-achieving schools, tracks, or classes, their academic self-concept often drops even if their own ability stays the same. This comparison cost shows up in selective schools, ability grouping, and stratified systems, and it appears across many countries and age groups. (18 sources)

A central empirical pattern is that school or class average achievement can predict *lower* academic self-concept once a student's own achievement is taken into account. This is the core comparison cost of the BFLPE: the same level of performance can feel strong in one setting and weak in another because students judge themselves against the local peer norm. Several papers describe this pattern directly in selective and high-performing contexts, including academically selective schools, high-ability classes, and high-track settings. (Gonzalez-Sauri et al., 2022) (Marsh et al., 2008) (Vasalampi et al., 2019) (Dinter et al., 2022) (Fang et al., 2018)

This is not just a theoretical claim. In a study of Anatolian high schools, higher school mathematics achievement predicted lower student mathematics self-concept, which the author interprets as evidence that social comparison in high-achieving schools shapes students' self-evaluations. The same paper argues that this matters even more in systems that sort students by academic capability, because grouping by achievement can strengthen the negative effect of school-average achievement on self-concept. (Kalaycioglu, 2017)

The same pattern is described in work on selective schools more broadly. Reviews and summaries note that high-performing students often develop lower self-concept when placed in selective schools than when they attend regular schools, even though their objective ability is unchanged. Meta-analytic evidence also points in this direction, estimating an average negative contextual effect of school or class achievement on academic self-concept. (Claro et al., 2022) (Fang et al., 2018) (Cecalupo et al., 2022) (Ahmed, 2025)

The evidence also extends to ability grouping and stratification inside school systems. Work on tracking and homogenous high-track classes suggests that ability stratification can increase BFLPE-type bias in children's academic self-concepts, because it places students in settings with stronger same-ability comparison targets. Studies discussing high-track classes note that even strong students can feel less capable when surrounded by equally strong or stronger classmates. (Parker et al., 2019) (Jonsdottir et al., 2022) (Webb-Williams, 2021) (Hoferichter et al., 2018)

This comparison cost is not limited to school-level academic sorting. Research on selective learning environments more generally finds that contexts filled with high-achieving peers can harm academic self-concept and, in some domains, motivation. Even advantageous neighborhood contexts may not help if they shift the comparison standard upward. (Fleischmann et al., 2021)

A notable feature of the evidence base is how broadly the pattern generalizes. One major review describes BFLPE findings as robust across settings, countries, follow-up periods, and research designs, with direct implications for tracking, ability grouping, selective schools, and gifted programs. A large meta-analysis likewise found a reliable negative average effect and suggested it can be stronger in high school samples. Cross-national work also reports that school-average achievement negatively predicts self-concept across many countries and across both higher- and lower-achieving students. (Marsh et al., 2008) (Fang et al., 2018) (Ahmed, 2025) (Marsh et al., 2019)

At the same time, the size of the cost can vary by transition and social context. For example, one study of the move into higher education did not find a clear BFLPE overall, suggesting that some students adapt well or that effects differ by field and performance level. More recent work also suggests the effect may be stronger for students who feel poorly emotionally integrated with peers, and less pronounced for those who are well

integrated socially and emotionally. This does not remove the basic pattern, but it does show that selective contexts do not affect all students in exactly the same way. (Gonzalez-Sauri et al., 2022) (Rosman et al., 2018) (Falk et al., 2024) (Ashkenazi et al., 2026)

Overall, the evidence supports a clear conclusion for the user's question: high-ability, selective, and grouped settings can carry real social comparison costs for academic self-concept. Those costs do not mean such settings are always harmful overall, but they do mean that "stronger peers" are not a simple benefit. For many students, especially in strongly stratified or selective environments, the price of being in a bigger pond is that they may start to see themselves as smaller. (Gonzalez-Sauri et al., 2022) (Marsh et al., 2008) (Jonsdottir et al., 2022) (Dinter et al., 2022)

Evidence

(Gonzalez-Sauri et al., 2022)

The Role of Early-Career University Prestige Stratification on the Future Academic Performance of Scholars

> *"The main hypothesis in this literature is the big-fish-little-pond effect (BFLPE). The hypothesis suggests that a student will have a lower academic self-concept (and thus performance) in an academically selective school, where the average achievement of peers is high, than in a non-selective one (Werts et al., 1969)(Marsh et al., 2003)(Marsh et al., 2008)(Salchegger, 2016)(Rosman et al., 2018)(Keyserlingk et al., 2020)."*

(Marsh et al., 2008)

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> *"The big-fish–little-pond effect (BFLPE) predicts that equally able students have lower academic self-concepts (ASCs) when attending schools where the average ability levels of classmates is high, and higher ASCs when attending schools where the school-average ability is low"*

> *".Because of the importance of ASC in predicting future achievement, coursework selection, and educational attainment, the results have important implications for the way in which schools are organized (e.g., tracking, ability grouping, academically selective schools, and gifted education programs)"*

> *".The flaw in this argument, as emphasized by (Hattie, 2002), is confounding the negative effects of ability grouping per se (the focus of the BFLPE) with the many other components that are likely to be incorporated into gifted education programs (e.g., different curriculums; more dedicated, highly trained teachers; better resources; enrichment experiences) that might be expected to have positive effects on ASC."*

(Vasalampi et al., 2019)

Classroom effect on primary school students' self-concept in literacy and mathematics

> *"It has been suggested that equally skilled students have lower self-concept of ability when they are placed in a classroom where the average performance level is high than when they attend a lowperforming classroom (Marsh, 1987)(Marsh et al., 2017). Thus, the same absolute level of performance can lead to different self-concept depending on the contexts in which students evaluate themselves. This phenomenon-known as the big-fish-little-pond effect (Marsh et al., 2007)(Marsh et al., 2008)-has been shown to be highly specific to the academic component of selfconcept. For example, (Marsh, 1987) demonstrated that there was a large negative classroom effect on academic self-concept but little or no negative classroom effect on general selfconcept or self-esteem"*

> *".The mechanism underlying the BFLP effect has been explained from a number of different perspectives (Marsh et al., 2010), which all emphasize that comparison processes are inevitable in a classroom (Marsh et al., 2007)(Marsh et al., 2017)."*

(Dinter et al., 2022)

The Effect of Fragile Self-Esteem on Course Completion in Higher Education

> *"The Big-Fish-Little-Pond-Effect (BFLPE) plays a special role in this context. It describes that students of the same ability in different academic environments come to different evaluations regarding their own academic self-concept and, thus, to different evaluations of the self: The higher the average academic level, the lower the academic self-concept (Marsh et al., 2008). Students who are located in an environment with other less able students than themselves may develop a higher academic self-concept and a higher self-esteem. On the other hand, the same students would develop a low academic self-concept if they were in an environment, in which there were more capable students than themselves. Thus, in terms of academic self-concept, it is more desirable to be a big fish in a small pond than the other way around"*

".This could lead to a negative evaluation of the self, rather than a more desirable assimilative effect (i.e. role model creation) based on an interpretive framework. Such an evaluation can be considered a shock according to the works of Köszegi et al. (2021). Given this background and assuming a high-ability environment, it would be reasonable to assume that the BPLFE exists and exerts an influence on students' academic self-concept. This influence would remove the objectivity of a comparison between assigned grades as representation of one's performance and would lead to decreased academic self-concept and, ultimately, self-esteem."

(Fang et al., 2018)

The Big-Fish-Little-Pond Effect on Academic Self-Concept: A Meta-Analysis

"The Big Fish Little Pond Effect (BFLPE) is one of the most influential theories about student ASC forming process, which was proposed by Marsh (1984) to describe the phenomenon that students in selective schools always have lower ASC compared to those with comparable ability but attend regular schools, which means that being a big fish in a small pond does good to one's ASC. Considerable evidence substantiated that the BFLPE is thought to be the outcome of individuals comparing their ability with the average ability of their group (Marsh, 1987;(Plieninger et al., 2015)"

".Students compare their own achievement with that of their class-or schoolmates, which leads them to feel more negative about their own competencies in high-achieving atmosphere than in low-achieving atmosphere."

(Kalaycioglu, 2017)

The Big Fish-Little Pond Effect on Affective Factors Based on PISA 2012 Mathematics Achievement

"The identification of increase in school's mathematics achievement yields a decrease in the student's mathematics self-concept may be considered as the most important result of this study. It is straightforward that social comparison at the Anatolian high schools, which have a high school mathematics achievement, has an important role on the students' self-evaluation. Thus, this result provides evidence about not only the Anatolian high school students but also the high achieving school students experience big fish-little pond effect, which is consistent with the international results (Marsh et al., 2014)(Marsh et al., 2003)(Nagengast et al., 2011)(Seaton et al., 2011)"

".Consequently, all high school students began to be grouped according to their academic capabilities. When the results of this study is evaluated by considering the new high school entrance system, the student's mathematics achievement is shown to be the most influential variable on the affective factors; however, especially, the school's mathematics achievement negative impact on the mathematics self-concept is undeniable."

(Claro et al., 2022)

Psychosocial reserve capacity, family background and selection of an educational path – a longitudinal study from Finland

"Additionally, tracking affects the self-concept or confidence in academic abilities of students through social comparison, known as the Big Fish Little Pond Effect (BFLPE). Current evidence has shown that high performing students develop lower self-concept when placed in selective schools (where other students also have high achievement), unlike in regular schools, where they have higher self-concept as a big fish in a little pond (Fang et al., 2018)Koivuhovi, 2021)."

(Cecalupo et al., 2022)

Individual Strivings in Social Comparison Processes: Achievement Motivation Goals in the Big-Fish-Little-Pond Effect

"Starting from these assumptions, it was theorized that equally able students end up having different views of their achievements depending on how high their peers' average level is (Marsh et al., 1984)(Marsh, 1987). Therefore, a student who attends a higher achieving school tends to have a lower concept of her/his academic achievement and school abilities, whereas a student who is embedded in a lower-achieving educational context ends up having a higher academic self-concept when comparing herself/himself to other students. Based on these hypotheses and on the results that the corpus of studies on the BFLPE has shown, it could be concluded that it is better to be a "big fish in a little pond" than a "little fish in a big pond." This means that, for a student, attending schools or educational programs in which the average achievement is lower or, at least, not higher than her/his own could be better."

(Ahmed, 2025)

Big fish in little ponds are less anxious about math: a multilevel analysis of school average achievement on math anxiety

"According to BFLPE, when students compare themselves to more able students, they would have lower self-concept; and when they compare themselves to less able students, they would have higher self-concept. For averageability students attending a high-achievement school can be a liability in terms of self-concept. According to the BFLPE, this educational context fosters upward social comparison which further leads to lower academic self-concepts. On the other hand, if these students attend low-achievement school, because this context fosters down-ward social comparison they would have higher academic self-concepts. This suggests that for a positive self-concept of ability it is better to be a big fish in a little pond. The BFLPE model holds that both individual students' achievements and the achievements of those in the school/classroom setting shape the students' self-concept of ability (Marsh, 1987)(Marsh et al., 2003). In support of BFLPE's predictions, a number of studies (e.g., (Marsh et al., 2019)(Marsh et al., 2021)Seaton et al., 2010; for a review see (Fang et al., 2018) have shown that school-average achievement negatively predicts students' self-concept of ability after controlling for individual achievement."

(Parker et al., 2019)

A macro context theory of academic self-concept: Ability stratification and the big-fish-little-pond effect

"ability stratification) will bias children's academic self-concepts via larger Big-Fish-Little-Pond effects (the negative effect of school average achievement on academic self-concept)."

(Jonsdottir et al., 2022)

The Choice of Track Matters: Academic Self-Concept and Sense of Purpose in Vocational and Academic Tracks

"Thus, the big-fish-little-pond effect occurs when strong students in high-ability schools compare themselves to their equally strong or stronger fellow students, causing them to feel less confident than if they would compare themselves to classmates in a low-ability school. Being a member of homogenous high-track class can thus have a negative impact on students' academic self-concept (Belfi et al., 2012)(Lavrijsen et al., 2017)(Marsh, 1987)(Trautwein et al., 2006)"

".The big-fish-little-pond effect has been hypothesized to be the net effect of the two counter-balancing processes that may occur concurrently: the positive assimilation effect of being a student in a highstatus class or school and the negative contrast effect of comparing oneself with academically strong (er) schoolmates. Most previous studies have concluded that the negative contrast effects are stronger than the positive assimilation effects (Belfi et al., 2012)(Marsh et al., 2000)(Preckel et al., 2010)(Trautwein et al., 2006)"

(Webb-Williams, 2021)

Teachers' Use of Within-Class Ability Groups in the Primary Classroom: A Mixed Methods Study of Social Comparison

"The Big Fish Little Pond Effect (BFLPE, (Marsh, 1984)(Marsh, 1987)Trautwein et al., 2009;Liem et al., 2013) has shown that students of comparable ability do better in lower-achieving classes or schools than in higher-achieving classes or schools. The understanding here is that children evaluate themselves more favorably against low-ability others than comparing to higher performing others which can lead to negative feelings about their own ability. Thus, it is better for a student's self-evaluation and academic self-concept to be a big fish in a large pond than a small fish in a big pond. Considerable evidence of this phenomenon has shown it to be robust internationally across gender and culture, arguing it be a result of social comparison (Marsh, 1987)"

".While BFLPE operates due to social comparisons, inconsistencies with Social Comparison Theory led researchers to call for direct evidence of the social comparison processes within BFLPE (Dai et al., 2008). Following the work by (Huguet et al., 2009) BFLPE was shown to be routed in how students compare with their school or class taken as a whole. School or class level comparisons in the BFLPE operate by using a generalized frame of reference where individuals make comparisons to the average/general ability of their class or school."

(Hoferichter et al., 2018)

The Big-Fish-Little-Pond Effect on the Four Facets of Academic Self-Concept

"the big-fish-little-pond-effect (BFLPE), according to which comparable talented students who attend high-ability schools have a lower ASC compared to equally talented students who attend comprehensive schools (Marsh, 1990)Seaton et al., 2009). Köller et al. (2006) explain this effect as follows: A student (little fish) who attends a highability class or school (big pond) has many opportunities for upward-comparisons, which in turn are related to a low selfconcept. However, if a student (big fish) would attend a rather low-ability class or school (small pond), the opportunity for downward-comparisons increase, leading to an increase of the student's self-concept."

(Fleischmann et al., 2021)

Living in the Big Pond: How Socioeconomic Neighborhood Composition Predicts Students' Academic Self-Concept

"Educational psychological research on the big fish little pond effect (BFLPE) suggests that selective learning environments, such as classrooms composed of high-achieving students, harm students' academic self-concept due to social comparison processes"

".Overall, we were unable to observe any positive effects of advantageous socioeconomic neighborhood conditions on students' academic self-concept. Instead, in specific domains, socioeconomic neighborhood conditions negatively predicted academic self-concept. In sum, our results suggest that advantageous socioeconomic neighborhood conditions might harm student motivation."

(Marsh et al., 2019)

Three Paradoxical Effects on Academic Self-Concept Across Countries, Schools, and Students

"Abstract. We simultaneously resolve three paradoxes in academic self-concept research with a single unifying meta-theoretical model based on frame-of-reference effects across 68 countries, 18,292 schools, and 485,490 15-year-old students. Paradoxically, but consistent with predictions, effects on math self-concepts were negative for: • being from countries where country-average achievement was high; explaining the paradoxical cross-cultural self-concept effect; • attending schools where school-average achievement was high; demonstrating big-fish-little-pond-effects (BFLPE) that generalized over 68 countries, Organisation for

Economic Co-operation and Development (OECD)/non-OECD countries, high/low achieving schools, and high/low achieving students; • year-in-school relative to age; unifying different research literatures for associated negative effects for starting school at a younger age and acceleration/skipping grades, and positive effects for starting school at an older age ("academic red shirting") and, paradoxically, even for repeating a grade. Contextual effects matter, resulting in significant and meaningful effects on self-beliefs, not only at the student (year in school) and local school level (BFLPE), but remarkably even at the macro-contextual country-level. Finally, we juxtapose cross-cultural generalizability based on Programme for International Student Assessment (PISA) data used here with generalizability based on meta-analyses, arguing that although the two approaches are similar in many ways, the generalizability shown here is stronger in terms of support for the universality of the frame-of-reference effects."

(Rosman et al., 2018)

Putting big fish into a bigger pond: self-concept changes in psychology undergraduate entrants

"ABSTRACT The transition to higher education is a topic of concern for educators around the world. In fact, the shift to a more competitive educational environment can negatively impact students' academic self-concept, which, in turn, might lead to lower persistence and less adaptive educational decisions. To examine how psychology students adapt to the transition to such an environment, the present article examines changes in psychology entrants' academic self-concepts. A four-wave longitudinal study covering students' first three semesters was conducted. Since the transition to higher education implies that students compare themselves to a stronger reference group, we expected academic self-concept to decrease over time (i.e. a big fish little pond effect). Moreover, we expected secondary school grades, study grades and gender to moderate self-concept development. While we found no evidence for a big fish little pond effect, we found that study grades and gender indeed moderate self-concept development: while self-concept scores of students with better grades increased with time, the contrary occurred for students with lower grades. Moreover, females' self-concepts remained largely constant whereas those of men increased considerably over time. We conclude that, in terms of self-concept changes, psychology undergraduates seem to adapt rather well to the transition to higher education. Future research should investigate whether or not these findings are generalisable to other academic fields, especially to disciplines that attract students with a broader range of achievements."

(Falk et al., 2024)

Self-Concept and Achievement in Individuals with Intellectual Disabilities

""The big-fish--little-pond effect (BFLPE), an application of social comparison theory to educational settings, posits that a student will have a lower academic self-concept in an academically selective school than in a nonselective school". (Marsh et al., 2003) (p. 364)"

"The strongest effect of the BFLPE was observed in students who were socially integrated (in school) but poorly emotionally integrated (with their peers). Conversely, the BFLPE was less pronounced in students who were well-integrated both in school and with their peers."

(Ashkenazi et al., 2026)

Recalibrating the 'average': gender differences in adolescents' normative social comparisons in schools of excellence

"According to the Big-Fish-Little-Pond Effect (BFLPE; (Marsh et al., 1984), equally able students tend to report lower academic self-concepts when surrounded by high-achieving peers, because self-evaluations are formed relative to the local achievement norm. In highly selective environments, even objectively strong students may therefore perceive themselves as merely average within the group"

". These dynamics may be particularly pronounced during the transition period, as students adjust to new academic standards and recalibrate their self-perceptions against a more demanding reference group."

Gifted students and specialized/gifted learning environments

Gifted students can face the same BFLPE comparison cost, and in some cases it may be especially visible because their reference group shifts from mixed-ability peers to other gifted peers. But a lower self-concept in gifted settings should not be treated as the whole story, because gifted programs can also provide better teaching, curriculum, and enrichment that may help students in other ways. (2 sources)

In gifted education, the BFLPE is often framed very directly: a gifted student may feel highly capable in a mixed-ability classroom but less capable after moving into a class or school filled with other gifted students, because the comparison standard rises. Lee et al. describe this as a contrast between learning with non-gifted peers and learning in specialized gifted settings, where gifted and talented students may define their ability and learner identity in relation to "superior peers." (Lee et al., 2020) This makes gifted programs a clear case of the user's concern about social comparison costs: the student's ability may stay high, but their academic self-concept can still drop when the pond gets more competitive. (Lee et al., 2020) (Marsh et al., 2008)

A notable point from the broader BFLPE literature is that gifted education should not be judged only by this comparison cost. Marsh et al. argue that the negative effect of being grouped with high-ability peers can be confused with the many other parts of gifted programs that may help students, such as different curricula, more trained teachers, better resources, and enrichment experiences. (Marsh et al., 2008) In other words, a gifted placement can lower academic self-concept through social comparison while still offering academic or developmental benefits through instruction and opportunity. (Marsh et al., 2008)

This is why gifted settings are especially important for interpretation. They show that “more able peers” are not a simple good for gifted students: such peers can provide challenge and stimulation, but they can also make students feel less exceptional, less confident, or less academically strong than they felt before. (Lee et al., 2020) (Marsh et al., 2008) For the user’s question, the key takeaway is that selective gifted environments can create real social comparison costs in academic self-concept, even when those same environments may also support learning in other ways. (Marsh et al., 2008)

Evidence

(Lee et al., 2020)

Risks of aiming to kill two birds with one stone: the affect of mathematically gifted and talented students in the dual realities of special schooling

> *"The big-fish-little-pond effect (BFLPE) refers to how G&T students demonstrate contrasting academic self-concepts in classrooms/ schools with non-gifted peers versus in specialized classrooms/schools with gifted peers. This notion is based on social comparative theory (see (Goethals et al., 1987) which posits that G&T students tend to define their ability and shape their learner identity in comparison with superior peers -that is, for some G&T students' academic self-concept, being a big "gifted" fish in a little pond with his or her "regular" or (not as gifted) peers is better."*

(Marsh et al., 2008)

The Big-fish–little-pond-effect Stands Up to Critical Scrutiny: Implications for Theory, Methodology, and Future Research

> *"The big-fish–little-pond effect (BFLPE) predicts that equally able students have lower academic self-concepts (ASCs) when attending schools where the average ability levels of classmates is high, and higher ASCs when attending schools where the school-average ability is low"*
>
> *".Because of the importance of ASC in predicting future achievement, coursework selection, and educational attainment, the results have important implications for the way in which schools are organized (e.g., tracking, ability grouping, academically selective schools, and gifted education programs)"*
>
> *".The flaw in this argument, as emphasized by (Hattie, 2002), is confounding the negative effects of ability grouping per se (the focus of the BFLPE) with the many other components that are likely to be incorporated into gifted education programs (e.g., different curriculums; more dedicated, highly trained teachers; better resources; enrichment experiences) that might be expected to have positive effects on ASC."*

Mechanisms: upward/downward comparison, contrast versus assimilation, and frames of reference

BFLPE works because students do not judge ability only by their own scores; they also use the achievement level of the people around them as a reference point. In most studies, upward comparison and contrast are stronger than any positive “I belong to a strong group” effect, so high-achieving contexts often lower academic self-concept. (17 sources)

The main mechanism behind BFLPE is social comparison inside a local frame of reference. Students look at their own achievement, but they also judge where they stand relative to classmates or schoolmates, so the same absolute performance can feel strong in one setting and weak in another. This is why equally able students can develop different academic self-concepts depending on the average achievement of their class or school. (Arens et al., 2015) (Hoffler et al., 2017) (Vasalampi et al., 2019) (Webb-Williams, 2021) (Jerrim et al., 2023) (Becker et al., 2024)

In simple terms, high-ability contexts increase the chance of upward comparison, while lower-ability contexts increase the chance of downward comparison. When students compare themselves with stronger peers, they

are more likely to lower their view of their own ability; when they compare themselves with weaker peers, they are more likely to feel capable and successful. This is the direct comparison-cost logic the user is asking about. (Hoferichter et al., 2018) (Basarkod et al., 2022) (Mai et al., 2022) (Ahmed, 2025) (Sabato et al., 2025)

A useful way to state this is as a contrast effect. The student compares the self to a generalized other—the class or school as a whole—and then sees a gap: "they are better than me" in a strong group, or "I am above average" in a weaker group. In BFLPE research, this contrast process is usually treated as the key psychological driver of lower academic self-concept in high-achieving environments. (Arens et al., 2015) (Webb-Williams, 2021) (Becker et al., 2024) (Fang et al., 2018)

Researchers also discuss a second process, assimilation. This is the idea that being in a high-status, high-performing class or school could lift self-views somewhat—through prestige, role models, or identification with a strong group. But the usual finding is that the negative contrast effect is larger, so the net result is still the classic BFLPE. (Jonsdottir et al., 2022) (Wu et al., 2023) (Jerrim et al., 2023) (Dinter et al., 2022)

This frame-of-reference account also helps explain why BFLPE is mostly about *academic* self-concept rather than self-worth in general. The comparisons are usually domain-specific: students compare math performance with other students' math performance, verbal performance with other students' verbal performance, and so on. Because the reference point is tied to school achievement, the strongest effects show up in students' beliefs about academic ability, not necessarily in global self-esteem. (Arens et al., 2015) (Vasalampi et al., 2019) (Mai et al., 2022)

Another useful detail is that these comparisons do not always require one clear rival. Students often compare themselves to the class average or to a general sense of "what students here are like," rather than to one specific classmate. That generalized comparison target helps explain why BFLPE can be strong in selective schools and tracked classes even when students are not openly competing with each other all the time. (Webb-Williams, 2021) (Becker et al., 2024)

Taken together, the mechanism is straightforward: students use the surrounding achievement level as a benchmark, high-achieving settings push many students toward upward comparison, and contrast usually outweighs assimilation. That is why stronger peer environments can create real comparison costs even for gifted or high-performing students. (Goetz et al., 2008) (Jonsdottir et al., 2022) (Wu et al., 2023) (Yang et al., 2024)

Evidence

(Arens et al., 2015)

How an early transition to high-ability secondary schools affects students' academic self-concept: Contrast effects, assimilation effects, and differential stability

> *"Contrast effects on students' academic self-concept have been investigated primarily in the context of research on the big-fish-little-pond effect (BFLPE; e.g., (Marsh, 1987). In this case, social comparison processes are assumed to yield differential effects on the individual (student) level and the group or context (class-average or school-average) level. Given that students compare their own achievement in one school subject with the achievement of other students in the same subject, high achievement leads to high levels of academic self-concept on the individual level. In addition, students compare their own achievement with the average achievement of the group to which they belong (within-group comparison; (Liu et al., 2005), leading to a negative effect of achievement on academic self-concept on the context level. Equally able students display low levels of academic self-concept in highability contexts, but high levels of academic self-concept in low-ability contexts. A student who is consistently confronted with better achieving students (i.e., with high class-average or school-average levels of achievement) might develop a poor academic self-concept, as this student always perceives his or her own accomplishments to be inferior to those of the other students. Conversely, the same student might perceive his or her accomplishments as above average in low-ability contexts, strengthening his or her academic self-concept."*

(Hoffler et al., 2017)

Science vs. Sports: Motivation and Self-concepts of Participants in Different School Competitions

> *"For instance, a well-established effect of social references is the big-fish-little-pond (BFLP) effect (Marsh, 1987): The mean*

achievement level of, for example, fellow students (the Bexternal frame of reference^; cf. internal/external frame of reference model, I/E model; (Marsh, 1986) can have a significant effect on the individual self-concept. Equally able students may develop different academic self-concepts depending on the academic abilities of their fellow students (the external frame of reference). If Bthe pond is small(i.e. the average ability level of the school's students is rather low), then an averageability student will use this reference for social comparison with the likely result that (s)he will develop a high academic self-concept (Marsh, 1987). If the same student was in a class with more able students than herself/himself, a significantly lower selfconcept would likely be the result (Marsh et al., 2008)."

(Vasalampi et al., 2019)

Classroom effect on primary school students' self-concept in literacy and mathematics

"It has been suggested that equally skilled students have lower self-concept of ability when they are placed in a classroom where the average performance level is high than when they attend a lowperforming classroom (Marsh, 1987)(Marsh et al., 2017). Thus, the same absolute level of performance can lead to different self-concept depending on the contexts in which students evaluate themselves. This phenomenon-known as the big-fish-little-pond effect (Marsh et al., 2007)(Marsh et al., 2008)-has been shown to be highly specific to the academic component of selfconcept. For example, (Marsh, 1987) demonstrated that there was a large negative classroom effect on academic self-concept but little or no negative classroom effect on general selfconcept or self-esteem"

".The mechanism underlying the BFLP effect has been explained from a number of different perspectives (Marsh et al., 2010), which all emphasize that comparison processes are inevitable in a classroom (Marsh et al., 2007)(Marsh et al., 2017)."

(Webb-Williams, 2021)

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"The Big Fish Little Pond Effect (BFLPE, (Marsh, 1984)(Marsh, 1987)Trautwein et al., 2009;Liem et al., 2013) has shown that students of comparable ability do better in lower-achieving classes or schools than in higher-achieving classes or schools. The understanding here is that children evaluate themselves more favorably against low-ability others than comparing to higher performing others which can lead to negative feelings about their own ability. Thus, it is better for a student's self-evaluation and academic self-concept to be a big fish in a large pond than a small fish in a big pond. Considerable evidence of this phenomenon has shown it to be robust internationally across gender and culture, arguing it be a result of social comparison (Marsh, 1987)"

".While BFLPE operates due to social comparisons, inconsistencies with Social Comparison Theory led researchers to call for direct evidence of the social comparison processes within BFLPE (Dai et al., 2008). Following the work by (Huguet et al., 2009) BFLPE was shown to be routed in how students compare with their school or class taken as a whole. School or class level comparisons in the BFLPE operate by using a generalized frame of reference where individuals make comparisons to the average/general ability of their class or school."

(Jerrim et al., 2023)

The big-fish-little-pond effect and overclaiming

"The Big-Fish-Little-Pond (BFLP) effect is a widely known psychological phenomena. It postulates that young people use their peers as a frame of reference to make inferences about their own academic abilities. Thus, one's relative ability in a subject in comparison to their school or class peers contributes to their own academic self-concept. The magnitude and direction of the association between one's relative academic position and their self-concept is driven by two countervailing forcesassimilation effects and contrast effects (Bless et al., 2016). While the former (assimilation effects) drives a young person's self-concept towards those of their peers over time, the latter (contrast effects) pushes it further away. Empirically, contrast effectsyoung people tending to become more confident when they are higher achieving than their peerstend to be the larger of the two (Wouters et al., 2013). This in-turn means that being one of the higher academic achievers within a school boosts a young person's academic self-belief, over and above their absolute level of academic achievement."

(Becker et al., 2024)

Living in the Big Pond: Adding the Neighborhood as a Frame of Reference for Academic Self-Concept Formation

"students' academic self-concept is negatively predicted by the average achievement of educational environments, that is, schools or classrooms when controlling for individual achievement (for an overview, see (Marsh et al., 2015). In other words, equally able students have lower academic self-concepts in high-achieving environments. This finding is called the bigfish-little-pond effect (BFLPE; (Marsh, 1987). The BFLPE emerges as a consequence of social comparison processes in which students evaluate their academic capabilities by comparing their achievement with that of other students in their educational environment (e.g., (Huguet et al., 2009)Marsh et al., 2014). The psychological mechanism causing the BFLPE is a contrast effect, implying that a student of given achievement forms contrasts with a generalized other-that is, the classroom or school environment-to fulfill comparison needs (Marsh et al., 2000)(Marsh et al., 2008). Thus, if a student assumes that this generalized other's achievement is superior (inferior), it harms (fosters) the student's self-concept."

(Hoferichter et al., 2018)

The Big-Fish-Little-Pond Effect on the Four Facets of Academic Self-Concept

"the big-fish-little-pond-effect (BFLPE), according to which comparable talented students who attend high-ability schools have a lower ASC compared to equally talented students who attend comprehensive schools (Marsh, 1990)Seaton et al., 2009). Köller et al. (2006) explain this effect as follows: A student (little fish) who attends a highability class or school (big pond) has many opportunities for upward-comparisons, which in turn are related to a low selfconcept. However, if a student (big fish) would attend a rather low-ability class or school (small pond), the opportunity for downward-comparisons increase, leading to an increase of the student's self-concept."

(Basarkod et al., 2022)

The Happy-Fish-Little-Pond Effect on Enjoyment: Generalizability Across Multiple Domains and Countries

"This is in line with the social comparison theory underlying the negative effect of group-average achievement on self-concept-the Big-Fish-Little-Pond Effect (Marsh et al., 1984). This theory posits that students evaluate their ability not only in relation to their own achievement levels, but also in relation to those around them. Thus, if a student is surrounded by peers who achieved top marks in a standardized test, that student is less likely to perceive their own performance as successful. In contrast, if a student is surrounded by peers who achieved low marks, the student will perceive their own score to be a success. Thus, higher class-average achievement is likely to lead to students feeling less in control of the subject, thereby lowering positive emotions and increasing negative emotions (Pekrun et al., 2019)."

(Mai et al., 2022)

Achievement Emotions in Selective Schools: Reexamining the Happy-Fish-Little-Pond Effect in an Extreme Case from the Chinese Collectivist Context

"The BFLPE describes the well-known phenomenon that students in selective schools/classes (big ponds) tend to have lower academic self-concepts than those who are comparably competent but attend regular schools/classes (little ponds), and this effect was shown to be stable and intercultural using considerable evidence (Marsh et al., 2003) and meta-analyses (Fang et al., 2018). The mechanism at the heart of these two effects lies in social comparison (Marsh, 1987). In addition to using objective, physical, and non-social criteria to evaluate their performance, individuals evaluate themselves by comparing themselves with others (Festinger, 1954). To some extent, feedback from the outside world is more important than objective self-evaluation standards (Klein, 1997). At school, peers who pursue similar or common goals are the main objects of comparison (Huang et al., 2019). According to the frame-of-reference effect, students' academic self-concepts and achievement emotions depend not only on their absolute academic level but also on their level relative to other students in the same school/class (Marsh, 1986). In selective schools/classes, being ranked low results in low academic self-concepts and more negative achievement emotions in students."

(Ahmed, 2025)

Big fish in little ponds are less anxious about math: a multilevel analysis of school average achievement on math anxiety

"According to BFLPE, when students compare themselves to more able students, they would have lower self-concept; and when they compare themselves to less able students, they would have higher self-concept. For averageability students attending a high-achievement school can be a liability in terms of self-concept. According to the BFLPE, this educational context fosters upward social comparison which further leads to lower academic self-concepts. On the other hand, if these students attend low-achievement school, because this context fosters down-ward social comparison they would have higher academic self-concepts. This suggests that for a positive self-concept of ability it is better to be a big fish in a little pond. The BFLPE model holds that both individual students' achievements and the achievements of those in the school/classroom setting shape the students' self-concept of ability (Marsh, 1987)(Marsh et al., 2003). In support of BFLPE's predictions, a number of studies (e.g., (Marsh et al., 2019)(Marsh et al., 2021)Seaton et al., 2010; for a review see (Fang et al., 2018) have shown that school-average achievement negatively predicts students' self-concept of ability after controlling for individual achievement."

(Sabato et al., 2025)

Being the Best, or With the Best: A Developmental Examination of Children's Choices in a Social Comparison Dilemma

"Examining the costs and benefits of upward versus downward comparison in terms of the students' self-concept, Marsh et al. (Marsh, 1987)(Marsh, 1993)Marsh andCraven 1997, 2002;(Marsh et al., 2003)(Marsh et al., 1984)(Marsh et al., 2014) consistently found a big fish in the little pond effect (BFLPE). According to their findings the cost of upward comparisons is embodied in a lower academic selfconcept, that students of equal ability are likely to exhibit in highachieving academic settings (vs. a higher academic self-concept of students in low-achieving environments, which involve fewer upward comparisons)."

(Fang et al., 2018)

The Big-Fish-Little-Pond Effect on Academic Self-Concept: A Meta-Analysis

"The Big Fish Little Pond Effect (BFLPE) is one of the most influential theories about student ASC forming process, which was proposed by Marsh (1984) to describe the phenomenon that students in selective schools always have lower ASC compared to those with comparable ability but attend regular schools, which means that being a big fish in a small pond does good to one's ASC.

Considerable evidence substantiated that the BFLPE is thought to be the outcome of individuals comparing their ability with the average ability of their group (Marsh, 1987;(Plieninger et al., 2015)"

".Students compare their own achievement with that of their class-or schoolmates, which leads them to feel more negative about their own competencies in high-achieving atmosphere than in low-achieving atmosphere."

(Jonsdottir et al., 2022)

The Choice of Track Matters: Academic Self-Concept and Sense of Purpose in Vocational and Academic Tracks

"Thus, the big-fish-little-pond effect occurs when strong students in high-ability schools compare themselves to their equally strong or stronger fellow students, causing them to feel less confident than if they would compare themselves to classmates in a low-ability school. Being a member of homogenous high-track class can thus have a negative impact on students' academic self-concept (Belfi et al., 2012)(Lavrijsen et al., 2017)(Marsh, 1987)(Trautwein et al., 2006)"

".The big-fish-little-pond effect has been hypothesized to be the net effect of the two counter-balancing processes that may occur concurrently: the positive assimilation effect of being a student in a highstatus class or school and the negative contrast effect of comparing oneself with academically strong (er) schoolmates. Most previous studies have concluded that the negative contrast effects are stronger than the positive assimilation effects (Belfi et al., 2012)(Marsh et al., 2000)(Preckel et al., 2010)(Trautwein et al., 2006)"

(Wu et al., 2023)

Association between School Contexts and the Development of Subjective Well-Being during Adolescence: A Context-Sensitive Longitudinal Study of Life Satisfaction and School Satisfaction

"Previous literature has found that achievement composition negatively predicted individual-level psychosocial outcomes, particularly academic self-belief (Becker et al., 2017)Frenzel et al., 2007;Marsh et al., 2008). This effect is known as the Big-Fish-Little-Pond effect, which can be viewed as the sum of contrast (negative) and assimilation (positive) effects. The contrast effect refers to situations in which students have more negative perceptions of themselves in high-ability groups than in low-ability groups through an upward social comparison process"

".When connecting the Big-Fish-Little-Pond effect and the construct of subjective well-being, one might assume that adolescents in academic tracks apply the upward social comparison to form their judgment of subjective well-being. If the upward social comparison (contrast effect) is stronger than the assimilation process (assimilation effect), adolescents in academic tracks should be less satisfied with their general and school lives."

(Dinter et al., 2022)

The Effect of Fragile Self-Esteem on Course Completion in Higher Education

"The Big-Fish-Little-Pond-Effect (BFLPE) plays a special role in this context. It describes that students of the same ability in different academic environments come to different evaluations regarding their own academic self-concept and, thus, to different evaluations of the self: The higher the average academic level, the lower the academic self-concept (Marsh et al., 2008). Students who are located in an environment with other less able students than themselves may develop a higher academic self-concept and a higher self-esteem. On the other hand, the same students would develop a low academic self-concept if they were in an environment, in which there were more capable students than themselves. Thus, in terms of academic self-concept, it is more desirable to be a big fish in a small pond than the other way around"

".This could lead to a negative evaluation of the self, rather than a more desirable assimilative effect (i.e. role model creation) based on an interpretive framework. Such an evaluation can be considered a shock according to the works of Köszegi et al. (2021). Given this background and assuming a high-ability environment, it would be reasonable to assume that the BPLFE exists and exerts an influence on students' academic self-concept. This influence would remove the objectivity of a comparison between assigned grades as representation of one's performance and would lead to decreased academic self-concept and, ultimately, self-esteem."

(Goetz et al., 2008)

Big fish in big ponds: A multilevel analysis of test anxiety and achievement in special gifted classes

"The BFLPE hypothesizes that it is better for academic self-concept, and, consequently, for test anxiety to be a big fish in a little pond (i.e., good student in a peer reference group of average ability), than to be a small fish in a big pond (i.e., good student in a peer reference group of high ability). (Marsh et al., 1984) proposed a social frame of reference model to explain the BFLPE. According to this model, self-perceptions in educational settings are largely shaped by the process of social comparison. With increasing ability level of a peer reference group (e.g., class), students often compare themselves with high ability peers and are compared by their teachers with more intellectually able peers, which, in turn, affects the feedback (e.g., grades) students receive."

(Yang et al., 2024)

Positive Attitudes Toward Mathematics and Big-Fish-Little-Pond Effects Among Eighth Grade Taiwanese Students

"While the major BFLPE studies and internal/external reference models (e.g., (Chiu, 2012) based on social comparison theory stated that when individuals evaluate their own abilities by comparing outstanding others, they will lower their self-concepts. Social comparisons may ultimately lead highachieving students, often referred to as "big fishes," to experience the "little-pond effect," where they perceive their abilities as less significant due to the lower average PATM levels at their schools."

Consequences beyond self-concept: emotions, motivation, values, and performance-related implications

The comparison cost of BFLPE does not stop at self-concept. In high-achieving settings, students can also feel more anxiety and other negative emotions, lose interest or motivation, value a subject less, and in some cases see weaker downstream performance-related outcomes. (12 sources)

The practical importance of BFLPE is that lower academic self-concept is tied to broader educational risks, not just to how students answer survey questions about ability. Work in this area explicitly links low academic self-concept with negative educational outcomes, which is why comparison costs in selective or grouped settings matter beyond self-image alone. (Jackman et al., 2011) A notable review also stresses that academic self-concept predicts later achievement, coursework selection, and educational attainment, making BFLPE relevant for students' longer-term academic pathways. (Marsh et al., 2008)

One clear downstream domain is emotion. BFLPE theory has been extended to test anxiety: it can be better not only for self-concept, but also for anxiety, to be a strong student in an average group than a strong student in a very high-ability group. Goetz et al. argue that as peer ability rises, students compare upward more often and may also receive more difficult comparative feedback from teachers, which can shape anxiety as well as self-beliefs. (Goetz et al., 2008) More recent work makes the emotional pattern even clearer: higher class-average achievement can make students feel less in control, lowering positive emotions and raising negative emotions. (Basarkod et al., 2022) (Pekrun et al., 2019) Similarly, students ranked low in selective schools or classes tend to report more negative achievement emotions. (Mai et al., 2022) (Fang et al., 2018)

These effects can reach beyond academic emotions into broader well-being. Wu et al. note that if upward comparison is stronger than any positive assimilation effect, adolescents in higher academic tracks may be less satisfied with both school life and life more generally. (Wu et al., 2023) Stockus et al. summarize the same concern in simple terms: students who are "little fish" in high-rank schools can show deflated self-evaluations, emotions, and aspirations. (Stockus et al., 2025) (Pekrun et al., 2019)

Motivation and task value can also be harmed. Fleischmann et al. report that advantageous contexts do not necessarily lift academic self-concept and may, in some domains, hurt student motivation. (Fleischmann et al., 2021) Newer summary work also points to lower perceived utility and intrinsic value of mathematics in higher-achieving classes, with self-concept acting as a mediator between class-average achievement and subject values. (Gholami et al., 2025) This matters for the user's concern about avoidance and retreat, because when students feel less capable and also value the subject less, they may become less willing to persist when work gets hard.

The performance story is more mixed, but it still raises concern. Some high-achieving settings may provide academic benefits, yet those gains do not reliably cancel out the emotional costs. Pekrun et al. found that any positive class-level effects on achievement were not consistent across studies, while the negative effects on emotion were more robust. (Pekrun et al., 2019) Stockus et al. therefore argue that deflated self-evaluations and aspirations in "little fish" students may contribute to lower performance and productivity. (Stockus et al., 2025) (Valentine et al., 2004)

Taken together, the evidence suggests that comparison costs can spill into several parts of school functioning at once: how students feel, how anxious they are, how much control they think they have, how much they value a subject, what they aspire to, and possibly how well they perform over time. That is why BFLPE should be understood not only as a self-concept effect, but as a broader risk process in selective, grouped, and high-pressure learning environments. (Jackman et al., 2011) (Marsh et al., 2008)

Evidence

(Jackman et al., 2011)

Big Fish in a Big Pond: a study of academic self concept in first year medical students

"Big-fish-little-pond effect (BFLPE) research has demonstrated that students in high-ability environments have lower academic self-concepts than equally able students in low-ability settings. Research has shown low academic self-concepts to be associated with negative educational outcomes. Social comparison processes have been implicated as fundamental to the BFLPE."

(Marsh et al., 2008)

The Big-fish–little-pond-effect Stands Up to Critical Scrutiny: Implications for Theory, Methodology, and Future Research

"The big-fish–little-pond effect (BFLPE) predicts that equally able students have lower academic self-concepts (ASCs) when attending schools where the average ability levels of classmates is high, and higher ASCs when attending schools where the school-average ability is low"

".Because of the importance of ASC in predicting future achievement, coursework selection, and educational attainment, the results have important implications for the way in which schools are organized (e.g., tracking, ability grouping, academically selective schools, and gifted education programs)"

".The flaw in this argument, as emphasized by (Hattie, 2002), is confounding the negative effects of ability grouping per se (the focus of the BFLPE) with the many other components that are likely to be incorporated into gifted education programs (e.g., different curriculums; more dedicated, highly trained teachers; better resources; enrichment experiences) that might be expected to have positive effects on ASC."

(Goetz et al., 2008)

Big fish in big ponds: A multilevel analysis of test anxiety and achievement in special gifted classes

"The BFLPE hypothesizes that it is better for academic self-concept, and, consequently, for test anxiety to be a big fish in a little pond (i.e., good student in a peer reference group of average ability), than to be a small fish in a big pond (i.e., good student in a peer reference group of high ability). (Marsh et al., 1984) proposed a social frame of reference model to explain the BFLPE. According to this model, self-perceptions in educational settings are largely shaped by the process of social comparison. With increasing ability level of a peer reference group (e.g., class), students often compare themselves with high ability peers and are compared by their teachers with more intellectually able peers, which, in turn, affects the feedback (e.g., grades) students receive."

(Basarkod et al., 2022)

The Happy-Fish-Little-Pond Effect on Enjoyment: Generalizability Across Multiple Domains and Countries

"This is in line with the social comparison theory underlying the negative effect of group-average achievement on self-concept-the Big-Fish-Little-Pond Effect (Marsh et al., 1984). This theory posits that students evaluate their ability not only in relation to their own achievement levels, but also in relation to those around them. Thus, if a student is surrounded by peers who achieved top marks in a standardized test, that student is less likely to perceive their own performance as successful. In contrast, if a student is surrounded by peers who achieved low marks, the student will perceive their own score to be a success. Thus, higher class-average achievement is likely to lead to students feeling less in control of the subject, thereby lowering positive emotions and increasing negative emotions (Pekrun et al., 2019)."

(Pekrun et al., 2019)

Happy fish in little ponds: Testing a reference group model of achievement and emotion.

"A theoretical model linking achievement and emotions is proposed. The model posits that individual achievement promotes positive achievement emotions and reduces negative achievement emotions. In contrast, group-level achievement is thought to reduce individuals' positive emotions and increase their negative emotions. The model was tested using one cross-sectional and two longitudinal datasets on 5th to 10th grade students' achievement emotions in mathematics (Studies 1-3: Ns = 1,610, 1,759, and 4,353, respectively). Multilevel latent structural equation modeling confirmed that individual achievement had positive predictive effects on positive emotions (enjoyment, pride) and negative predictive effects on negative emotions (anger, anxiety, shame, and hopelessness), controlling for prior achievement, autoregressive effects, reciprocal effects, gender, and socioeconomic status (SES). Class-level achievement had negative compositional effects on the positive emotions and positive compositional effects on the negative emotions. Additional analyses suggested that self-concept of ability is a possible mediator of these effects. Furthermore, there were positive compositional effects of class-level achievement on individual achievement in Study 2 but not in Study 3, indicating that negative compositional effects on emotion are not reliably counteracted by positive effects on performance. The results were robust across studies, age groups, synchronous versus longitudinal analysis, and latent-manifest versus doubly latent modeling. These findings imply that individual success drives emotional well-being, whereas placing individuals in high-achieving groups can undermine well-being. Thus, the findings challenge policy and practice decisions on achievement-contingent allocation

of individuals to groups. (PsycINFO Database Record (c) 2019 APA, all rights reserved)."

(Mai et al., 2022)

Achievement Emotions in Selective Schools: Reexamining the Happy-Fish-Little-Pond Effect in an Extreme Case from the Chinese Collectivist Context

"The BFLPE describes the well-known phenomenon that students in selective schools/classes (big ponds) tend to have lower academic self-concepts than those who are comparably competent but attend regular schools/classes (little ponds), and this effect was shown to be stable and intercultural using considerable evidence (Marsh et al., 2003) and meta-analyses (Fang et al., 2018). The mechanism at the heart of these two effects lies in social comparison (Marsh, 1987). In addition to using objective, physical, and non-social criteria to evaluate their performance, individuals evaluate themselves by comparing themselves with others (Festinger, 1954). To some extent, feedback from the outside world is more important than objective self-evaluation standards (Klein, 1997). At school, peers who pursue similar or common goals are the main objects of comparison (Huang et al., 2019). According to the frame-of-reference effect, students' academic self-concepts and achievement emotions depend not only on their absolute academic level but also on their level relative to other students in the same school/class (Marsh, 1986). In selective schools/classes, being ranked low results in low academic self-concepts and more negative achievement emotions in students."

(Fang et al., 2018)

The Big-Fish-Little-Pond Effect on Academic Self-Concept: A Meta-Analysis

"The Big Fish Little Pond Effect (BFLPE) is one of the most influential theories about student ASC forming process, which was proposed by Marsh (1984) to describe the phenomenon that students in selective schools always have lower ASC compared to those with comparable ability but attend regular schools, which means that being a big fish in a small pond does good to one's ASC. Considerable evidence substantiated that the BFLPE is thought to be the outcome of individuals comparing their ability with the average ability of their group (Marsh, 1987;(Plieninger et al., 2015)"

".Students compare their own achievement with that of their class-or schoolmates, which leads them to feel more negative about their own competencies in high-achieving atmosphere than in low-achieving atmosphere."

(Wu et al., 2023)

Association between School Contexts and the Development of Subjective Well-Being during Adolescence: A Context-Sensitive Longitudinal Study of Life Satisfaction and School Satisfaction

"Previous literature has found that achievement composition negatively predicted individual-level psychosocial outcomes, particularly academic self-belief (Becker et al., 2017)Frenzel et al., 2007;Marsh et al., 2008). This effect is known as the Big-Fish-Little-Pond effect, which can be viewed as the sum of contrast (negative) and assimilation (positive) effects. The contrast effect refers to situations in which students have more negative perceptions of themselves in high-ability groups than in low-ability groups through an upward social comparison process"

".When connecting the Big-Fish-Little-Pond effect and the construct of subjective well-being, one might assume that adolescents in academic tracks apply the upward social comparison to form their judgment of subjective well-being. If the upward social comparison (contrast effect) is stronger than the assimilation process (assimilation effect), adolescents in academic tracks should be less satisfied with their general and school lives."

(Stockus et al., 2025)

Do growth mindsets reduce the big-fish-little-pond effect?

"Although the BFLPE is beneficial for students with high rank in low rank schools, it is detrimental for students with low rank in high rank schools. That is, students who are little fish in big ponds experience deflated self-evaluations, emotions, and aspirations (Nagengast et al., 2012)(Pekrun et al., 2019), which may contribute to decrements in performance and productivity (Marsh et al., 2011)(Valentine et al., 2004). Thus, it is important to identify factors that reduce the harmful effects of being a little fish in a big pond. Along these lines, scholars have recently noted that there is a need to "more fully recognize the negative implications of the BFLPE and to pursue research evaluating strategies designed to counter the negative effects" (Marsh et al., 2019, p. 237)."

(Fleischmann et al., 2021)

Living in the Big Pond: How Socioeconomic Neighborhood Composition Predicts Students' Academic Self-Concept

"Educational psychological research on the big fish little pond effect (BFLPE) suggests that selective learning environments, such as classrooms composed of high-achieving students, harm students' academic self-concept due to social comparison processes"

".Overall, we were unable to observe any positive effects of advantageous socioeconomic neighborhood conditions on students' academic self-concept. Instead, in specific domains, socioeconomic neighborhood conditions negatively predicted academic self-concept. In sum, our results suggest that advantageous socioeconomic neighborhood conditions might harm student motivation."

(Gholami et al., 2025)

The Big-Fish-Little-Pond effect on mathematics students' intelligence mindset in selective and non-selective schools

> *"This effect arises from social comparison processes and has been documented across diverse student populations and contexts, making it one of the most replicated findings in educational psychology (Marsh et al., 2019)(Stockus et al., 2023)"*
>
> *".Recent evidence shows that the BFLPE influences not only general academic self-concept but also subject-specific domains such as mathematics, language, and reading"*
>
> *".Similarly, research in Germany found that the BFLPE negatively affected students' perceptions of the utility and intrinsic value of mathematics, with self-concept mediating the relationship between class-average achievement and academic values (Cambria et al., 2017). The effect also extends beyond cognitive domains: in a study of 7,000 Chinese adolescents, higher school-average mathematics achievement was associated with lower levels of nine key social-emotional skills, including self-efficacy, responsibility, and persistence (Shi et al., 2025)."*

(Valentine et al., 2004)

The Relation Between Self-Beliefs and Academic Achievement: A Meta-Analytic Review

> *"There has been extensive debate among scholars and practitioners concerning whether self-beliefs influence academic achievement. To address this question, findings of longitudinal studies investigating the relation between self-beliefs and achievement were synthesized using meta-analysis. Estimated effects are consistent with a small, favorable influence of positive self-beliefs on academic achievement, with an average standardized path or regression coefficient of .08 for self-beliefs as a predictor of later achievement, controlling for initial levels of achievement. Stronger effects of self-beliefs are evident when assessing self-beliefs specific to the academic domain and when measures of self-beliefs and achievement are matched by domain (e.g., same subject area). Under these conditions, the relation of self-beliefs to later achievement meets or exceeds Cohen's (1988) definition of a small effect size."*

Implications for school organization and transitions

BFLPE matters for how schools sort students. Tracking, selective admissions, and high-track grouping can raise standards and opportunities, but they can also lower students' academic self-concept by shifting the comparison group upward. (10 sources)

A main implication of the BFLPE literature is that school organization is not neutral for student self-beliefs. When systems use tracking, ability grouping, selective schools, or gifted programs, they do not just place students into different instructional settings; they also change the peer norm students use to judge their own ability. Because academic self-concept helps shape later achievement, course choice, and attainment, this gives BFLPE direct relevance for policy and school design. (Marsh et al., 2008) (Gonzalez-Sauri et al., 2022)

This is especially important in systems that sort students early or strongly by achievement. Research on ability stratification argues that such sorting can bias children's academic self-concepts by producing larger BFLPEs, and work on tracking similarly notes that high-performing students often report lower self-concept in selective schools than in regular schools because they are no longer the "big fish." (Parker et al., 2019) (Claro et al., 2022) (Fang et al., 2018) High-track and homogenous classes therefore pose a practical tradeoff for schools: they may offer challenge and status, but they also place students into stronger comparison fields where contrast tends to outweigh any positive assimilation effect. (Jonsdottir et al., 2022)

The policy point is not that all grouping is bad, but that grouping decisions come with psychological costs that schools should expect and plan for. Marsh et al. make this point clearly, arguing that BFLPE has implications for tracking, ability grouping, academically selective schools, and gifted education, while also warning against confusing the negative comparison effect of grouping itself with the possible benefits of better curriculum, teaching, resources, or enrichment in special programs. (Marsh et al., 2008) In the same way, Kalaycioglu argues that when high school systems group students by academic capability, the negative effect of school-average mathematics achievement on mathematics self-concept becomes hard to ignore. (Kalaycioglu, 2017)

Transitions are another key implication. Any move into a more selective setting—a higher track, a selective school, a gifted class, or university—can force students to recalibrate their self-view against a stronger local norm. This is why transition periods are often seen as times when comparison costs may be especially strong. (Ashkenazi et al., 2026) (Gonzalez-Sauri et al., 2022) (Marsh et al., 2008) Still, transition effects are

not uniform. One higher-education study found no overall BFLPE and suggested that some students adapt well, while students with weaker grades may show declining self-concept over time. (Gonzalez-Sauri et al., 2022) (Rosman et al., 2018)

This suggests a practical lesson for schools: the risk is not only admission into a selective setting, but also how students are supported after entry. Recent evidence indicates that the BFLPE can be stronger for students who are poorly emotionally integrated with peers and weaker for those who are well integrated both socially and emotionally. (Falk et al., 2024) So if schools use selective grouping, transition supports should not focus only on academic adjustment. They should also help students interpret rank changes, normalize the shift from being top of the class to being one among many strong students, and build peer belonging in ways that reduce harmful comparison pressure. (Falk et al., 2024) (Ashkenazi et al., 2026)

For the user's question, the bottom line is simple: BFLPE implies that school structure can create comparison costs. The more a system sorts students into high-ability ponds, especially at key transitions, the more it should expect some students to feel smaller—even when their actual ability has not changed. (Marsh et al., 2008) (Parker et al., 2019) (Jonsdottir et al., 2022)

Evidence

(Marsh et al., 2008)

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> *"The big-fish–little-pond effect (BFLPE) predicts that equally able students have lower academic self-concepts (ASCs) when attending schools where the average ability levels of classmates is high, and higher ASCs when attending schools where the school-average ability is low"*
>
> *".Because of the importance of ASC in predicting future achievement, coursework selection, and educational attainment, the results have important implications for the way in which schools are organized (e.g., tracking, ability grouping, academically selective schools, and gifted education programs)"*
>
> *".The flaw in this argument, as emphasized by (Hattie, 2002), is confounding the negative effects of ability grouping per se (the focus of the BFLPE) with the many other components that are likely to be incorporated into gifted education programs (e.g., different curriculums; more dedicated, highly trained teachers; better resources; enrichment experiences) that might be expected to have positive effects on ASC."*

(Gonzalez-Sauri et al., 2022)

The Role of Early-Career University Prestige Stratification on the Future Academic Performance of Scholars

> *"The main hypothesis in this literature is the big-fish-little-pond effect (BFLPE). The hypothesis suggests that a student will have a lower academic self-concept (and thus performance) in an academically selective school, where the average achievement of peers is high, than in a non-selective one (Werts et al., 1969)(Marsh et al., 2003)(Marsh et al., 2008)(Salchegger, 2016)(Rosman et al., 2018)(Keyserlingk et al., 2020)."*

(Parker et al., 2019)

A macro context theory of academic self-concept: Ability stratification and the big-fish-little-pond effect

> *"ability stratification) will bias children's academic self-concepts via larger Big-Fish-Little-Pond effects (the negative effect of school average achievement on academic self-concept)."*

(Claro et al., 2022)

Psychosocial reserve capacity, family background and selection of an educational path – a longitudinal study from Finland

> *"Additionally, tracking affects the self-concept or confidence in academic abilities of students through social comparison, known as the Big Fish Little Pond Effect (BFLPE). Current evidence has shown that high performing students develop lower self-concept when placed in selective schools (where other students also have high achievement), unlike in regular schools, where they have higher self-concept as a big fish in a little pond (Fang et al., 2018)Koivuhovi, 2021)."*

(Fang et al., 2018)

The Big-Fish-Little-Pond Effect on Academic Self-Concept: A Meta-Analysis

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proposed by Marsh (1984) to describe the phenomenon that students in selective schools always have lower ASC compared to those with comparable ability but attend regular schools, which means that being a big fish in a small pond does good to one's ASC. Considerable evidence substantiated that the BFLPE is thought to be the outcome of individuals comparing their ability with the average ability of their group (Marsh, 1987;(Plieninger et al., 2015)"

".Students compare their own achievement with that of their class-or schoolmates, which leads them to feel more negative about their own competencies in high-achieving atmosphere than in low-achieving atmosphere."

(Jonsdottir et al., 2022)

The Choice of Track Matters: Academic Self-Concept and Sense of Purpose in Vocational and Academic Tracks

"Thus, the big-fish-little-pond effect occurs when strong students in high-ability schools compare themselves to their equally strong or stronger fellow students, causing them to feel less confident than if they would compare themselves to classmates in a low-ability school. Being a member of homogenous high-track class can thus have a negative impact on students' academic self-concept (Belfi et al., 2012)(Lavrijsen et al., 2017)(Marsh, 1987)(Trautwein et al., 2006)"

".The big-fish-little-pond effect has been hypothesized to be the net effect of the two counter-balancing processes that may occur concurrently: the positive assimilation effect of being a student in a highstatus class or school and the negative contrast effect of comparing oneself with academically strong (er) schoolmates. Most previous studies have concluded that the negative contrast effects are stronger than the positive assimilation effects (Belfi et al., 2012)(Marsh et al., 2000)(Preckel et al., 2010)(Trautwein et al., 2006)"

(Kalaycioglu, 2017)

The Big Fish-Little Pond Effect on Affective Factors Based on PISA 2012 Mathematics Achievement

"The identification of increase in school's mathematics achievement yields a decrease in the student's mathematics self-concept may be considered as the most important result of this study. It is straightforward that social comparison at the Anatolian high schools, which have a high school mathematics achievement, has an important role on the students' self-evaluation. Thus, this result provides evidence about not only the Anatolian high school students but also the high achieving school students experience big fish-little pond effect, which is consistent with the international results (Marsh et al., 2014)(Marsh et al., 2003)(Nagengast et al., 2011)(Seaton et al., 2011)"

".Consequently, all high school students began to be grouped according to their academic capabilities. When the results of this study is evaluated by considering the new high school entrance system, the student's mathematics achievement is shown to be the most influential variable on the affective factors; however, especially, the school's mathematics achievement negative impact on the mathematics self-concept is undeniable."

(Ashkenazi et al., 2026)

Recalibrating the ‘average’: gender differences in adolescents’ normative social comparisons in schools of excellence

"According to the Big-Fish-Little-Pond Effect (BFLPE; (Marsh et al., 1984), equally able students tend to report lower academic self-concepts when surrounded by high-achieving peers, because self-evaluations are formed relative to the local achievement norm. In highly selective environments, even objectively strong students may therefore perceive themselves as merely average within the group"

". These dynamics may be particularly pronounced during the transition period, as students adjust to new academic standards and recalibrate their self-perceptions against a more demanding reference group."

(Rosman et al., 2018)

Putting big fish into a bigger pond: self-concept changes in psychology undergraduate entrants

"ABSTRACT The transition to higher education is a topic of concern for educators around the world. In fact, the shift to a more competitive educational environment can negatively impact students’ academic self-concept, which, in turn, might lead to lower persistence and less adaptive educational decisions. To examine how psychology students adapt to the transition to such an environment, the present article examines changes in psychology entrants’ academic self-concepts. A four-wave longitudinal study covering students’ first three semesters was conducted. Since the transition to higher education implies that students compare themselves to a stronger reference group, we expected academic self-concept to decrease over time (i.e. a big fish little pond effect). Moreover, we expected secondary school grades, study grades and gender to moderate self-concept development. While we found no evidence for a big fish little pond effect, we found that study grades and gender indeed moderate self-concept development: while self-concept scores of students with better grades increased with time, the contrary occurred for students with lower grades. Moreover, females’ self-concepts remained largely constant whereas those of men increased considerably over time. We conclude that, in terms of self-concept changes, psychology undergraduates seem to adapt rather well to the transition to higher education. Future research should investigate whether or not these findings are generalisable to other academic fields, especially to disciplines that attract students with a broader range of achievements."

(Falk et al., 2024)

Self-Concept and Achievement in Individuals with Intellectual Disabilities

> *""The big-fish--little-pond effect (BFLPE), an application of social comparison theory to educational settings, posits that a student will have a lower academic self-concept in an academically selective school than in a nonselective school". (Marsh et al., 2003) (p. 364)"*
>
> *"The strongest effect of the BFLPE was observed in students who were socially integrated (in school) but poorly emotionally integrated (with their peers). Conversely, the BFLPE was less pronounced in students who were well-integrated both in school and with their peers."*

Behavioral retreat, risky learning, and early avoidance

BFLPE can matter not just for how students feel, but for what they do. When strong students feel smaller in a high-ability setting, they can become less willing to take learning risks, less persistent, and more likely to pull back early from challenge. (12 sources)

A useful way to connect BFLPE to behavior is to start with rank. In selective classes and schools, students who end up low in the local ranking often report lower academic self-concept and more negative achievement emotions, even when they are objectively competent. That combination can make demanding work feel less controllable and less worth the risk, which fits the idea of behavioral retreat: students may protect themselves by lowering effort, avoiding challenge, or stepping back from situations where failure could become visible. (Mai et al., 2022) (Fang et al., 2018) (Basarkod et al., 2022) (Pekrun et al., 2019)

The evidence is stronger for this chain than for any single behavior label. Studies show that higher-achieving classes can reduce positive emotions and increase negative emotions, with self-concept as a likely mediator, and those emotional shifts are the kinds of conditions under which students often become more cautious and less exploratory in learning. In plain terms, if a student starts to feel “I am not good at this here,” risky learning becomes harder: answering in public, trying a harder problem, choosing an advanced course, or persisting through confusion all carry more self-threat. (Basarkod et al., 2022) (Pekrun et al., 2019) (Stockus et al., 2025)

There is also direct reason to worry about motivation and value loss as pathways to early avoidance. Fleischmann et al. found that selective or otherwise advantageous contexts did not boost self-concept and in some domains could harm motivation. Related work summarized by Gholami et al. reports lower utility and intrinsic value for mathematics in higher-achieving classes, with self-concept mediating the link between class-average achievement and academic values. If students both doubt their ability and value the subject less, retreat from challenge becomes more likely. (Fleischmann et al., 2021) (Gholami et al., 2025)

This helps make sense of the user’s idea of “risky learning.” High-ability environments can shift students away from mastery and toward position-monitoring: not just “Can I learn this?” but “How do I look relative to others?” That kind of comparison focus can discourage open error, experimentation, and ambitious choices, especially for students who are no longer near the top of the group. Mai et al. note that peers in school are the main comparison targets, and Huang et al.’s work on goal pursuit shows how relative position can crowd out objective progress and even reduce later effort after competitive gains. While this is not a direct BFLPE study of classroom avoidance, it supports the idea that strong comparison climates can pull behavior away from learning and toward self-protection. (Mai et al., 2022) (Huang et al., 2019)

The likely downstream pattern is therefore cumulative. Students who become “little fish” can show deflated self-evaluations, emotions, and aspirations, and these may contribute to lower performance and productivity over time. Because academic self-beliefs have a small but reliable effect on later achievement, repeated comparison costs can plausibly feed an early-avoidance cycle: lower self-concept, less confidence in control, lower value, less willingness to take on challenge, and then fewer chances to build competence through difficult work. (Stockus et al., 2025) (Valentine et al., 2004)

So the best careful conclusion is not that BFLPE has already proven one single behavior such as dropout, withdrawal, or avoidance in every setting. Rather, the literature shows a cluster of risks that point in that

direction: lower self-concept, less control, more negative emotion, weaker motivation, lower value, and reduced aspirations in high-rank environments. For gifted students or students in selective groups, that means the social comparison cost can show up as an early retreat from hard, uncertain learning precisely when challenge is supposed to help them grow. (Marsh et al., 2008) (Wu et al., 2023) (Marsh et al., 2019)

Evidence

(Mai et al., 2022)

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(Basarkod et al., 2022)

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(Pekrun et al., 2019)

Happy fish in little ponds: Testing a reference group model of achievement and emotion.

> *"A theoretical model linking achievement and emotions is proposed. The model posits that individual achievement promotes positive achievement emotions and reduces negative achievement emotions. In contrast, group-level achievement is thought to reduce individuals' positive emotions and increase their negative emotions. The model was tested using one cross-sectional and two longitudinal datasets on 5th to 10th grade students' achievement emotions in mathematics (Studies 1-3: Ns = 1,610, 1,759, and 4,353, respectively). Multilevel latent structural equation modeling confirmed that individual achievement had positive predictive effects on positive emotions (enjoyment, pride) and negative predictive effects on negative emotions (anger, anxiety, shame, and hopelessness), controlling for prior achievement, autoregressive effects, reciprocal effects, gender, and socioeconomic status (SES). Class-level achievement had negative compositional effects on the positive emotions and positive compositional effects on the negative emotions. Additional analyses suggested that self-concept of ability is a possible mediator of these effects. Furthermore, there were positive compositional effects of class-level achievement on individual achievement in Study 2 but not in Study 3, indicating that negative compositional effects on emotion are not reliably counteracted by positive effects on performance. The results were robust across studies, age groups, synchronous versus longitudinal analysis, and latent-manifest versus doubly latent modeling. These findings imply that individual success drives emotional well-being, whereas placing individuals in high-achieving groups can undermine well-being. Thus, the findings challenge policy and practice decisions on achievement-contingent allocation*

of individuals to groups. (PsycINFO Database Record (c) 2019 APA, all rights reserved)."

(Stockus et al., 2025)

Do growth mindsets reduce the big-fish-little-pond effect?

"Although the BFLPE is beneficial for students with high rank in low rank schools, it is detrimental for students with low rank in high rank schools. That is, students who are little fish in big ponds experience deflated self-evaluations, emotions, and aspirations (Nagengast et al., 2012)(Pekrun et al., 2019), which may contribute to decrements in performance and productivity (Marsh et al., 2011)(Valentine et al., 2004). Thus, it is important to identify factors that reduce the harmful effects of being a little fish in a big pond. Along these lines, scholars have recently noted that there is a need to "more fully recognize the negative implications of the BFLPE and to pursue research evaluating strategies designed to counter the negative effects" (Marsh et al., 2019, p. 237)."

(Fleischmann et al., 2021)

Living in the Big Pond: How Socioeconomic Neighborhood Composition Predicts Students' Academic Self-Concept

"Educational psychological research on the big fish little pond effect (BFLPE) suggests that selective learning environments, such as classrooms composed of high-achieving students, harm students' academic self-concept due to social comparison processes"

".Overall, we were unable to observe any positive effects of advantageous socioeconomic neighborhood conditions on students' academic self-concept. Instead, in specific domains, socioeconomic neighborhood conditions negatively predicted academic self-concept. In sum, our results suggest that advantageous socioeconomic neighborhood conditions might harm student motivation."

(Gholami et al., 2025)

The Big-Fish-Little-Pond effect on mathematics students' intelligence mindset in selective and non-selective schools

"This effect arises from social comparison processes and has been documented across diverse student populations and contexts, making it one of the most replicated findings in educational psychology (Marsh et al., 2019)(Stockus et al., 2023)"

".Recent evidence shows that the BFLPE influences not only general academic self-concept but also subject-specific domains such as mathematics, language, and reading"

".Similarly, research in Germany found that the BFLPE negatively affected students' perceptions of the utility and intrinsic value of mathematics, with self-concept mediating the relationship between class-average achievement and academic values (Cambria et al., 2017). The effect also extends beyond cognitive domains: in a study of 7,000 Chinese adolescents, higher school-average mathematics achievement was associated with lower levels of nine key social-emotional skills, including self-efficacy, responsibility, and persistence (Shi et al., 2025)."

(Huang et al., 2019)

When individual goal pursuit turns competitive: How we sabotage and coast.

"People working toward individual goals often find themselves surrounded by others who are pursuing similar goals, such as at school, in fitness classes, and through goal-oriented network devices like Fitbit. This research explores when these individual goal pursuits can turn into competitions, why it happens, and the downstream consequences of this pseudocompetition on goal pursuers. We found that people were more likely to treat their goal pursuit as a competition when they were near the end (vs. at the beginning) of their individual goal and, thus, prioritized relative positional gain (i.e., performing better than others sharing similar pursuits) over making objective progress on their own goal, sabotaging others when they had the opportunity to do so (Studies 1-3B). Further, we provided evidence that certainty of goal attainment at a high (vs. low) level of progress drove this shift in focus, leading to such sabotage behaviors (Studies 3A and 3B). Ironically, success in gaining an upper hand against others in these pseudocompetitions led individuals to subsequently reduce their effort in their own pursuits (Studies 1-5). Six experiments captured a variety of competitive behaviors across different goal domains (e.g., selecting games that diminished others' prospects, selecting difficult questions for fellow students). (PsycINFO Database Record (c) 2019 APA, all rights reserved)."

(Valentine et al., 2004)

The Relation Between Self-Beliefs and Academic Achievement: A Meta-Analytic Review

"There has been extensive debate among scholars and practitioners concerning whether self-beliefs influence academic achievement. To address this question, findings of longitudinal studies investigating the relation between self-beliefs and achievement were synthesized using meta-analysis. Estimated effects are consistent with a small, favorable influence of positive self-beliefs on academic achievement, with an average standardized path or regression coefficient of .08 for self-beliefs as a predictor of later achievement, controlling for initial levels of achievement. Stronger effects of self-beliefs are evident when assessing self-beliefs specific to the academic domain and when measures of self-beliefs and achievement are matched by domain (e.g., same subject area). Under these conditions, the relation of self-beliefs to later achievement meets or exceeds Cohen's (1988) definition of a small effect size."

(Marsh et al., 2008)

The Big-fish–little-pond-effect Stands Up to Critical Scrutiny: Implications for Theory, Methodology, and Future Research

> *"The big-fish–little-pond effect (BFLPE) predicts that equally able students have lower academic self-concepts (ASCs) when attending schools where the average ability levels of classmates is high, and higher ASCs when attending schools where the school-average ability is low"*
>
> *".Because of the importance of ASC in predicting future achievement, coursework selection, and educational attainment, the results have important implications for the way in which schools are organized (e.g., tracking, ability grouping, academically selective schools, and gifted education programs)"*
>
> *".The flaw in this argument, as emphasized by (Hattie, 2002), is confounding the negative effects of ability grouping per se (the focus of the BFLPE) with the many other components that are likely to be incorporated into gifted education programs (e.g., different curriculums; more dedicated, highly trained teachers; better resources; enrichment experiences) that might be expected to have positive effects on ASC."*

(Wu et al., 2023)

Association between School Contexts and the Development of Subjective Well-Being during Adolescence: A Context-Sensitive Longitudinal Study of Life Satisfaction and School Satisfaction

> *"Previous literature has found that achievement composition negatively predicted individual-level psychosocial outcomes, particularly academic self-belief (Becker et al., 2017)Frenzel et al., 2007;Marsh et al., 2008). This effect is known as the Big-Fish-Little-Pond effect, which can be viewed as the sum of contrast (negative) and assimilation (positive) effects. The contrast effect refers to situations in which students have more negative perceptions of themselves in high-ability groups than in low-ability groups through an upward social comparison process"*
>
> *".When connecting the Big-Fish-Little-Pond effect and the construct of subjective well-being, one might assume that adolescents in academic tracks apply the upward social comparison to form their judgment of subjective well-being. If the upward social comparison (contrast effect) is stronger than the assimilation process (assimilation effect), adolescents in academic tracks should be less satisfied with their general and school lives."*

(Marsh et al., 2019)

Three Paradoxical Effects on Academic Self-Concept Across Countries, Schools, and Students

> *"Abstract. We simultaneously resolve three paradoxes in academic self-concept research with a single unifying meta-theoretical model based on frame-of-reference effects across 68 countries, 18,292 schools, and 485,490 15-year-old students. Paradoxically, but consistent with predictions, effects on math self-concepts were negative for: • being from countries where country-average achievement was high; explaining the paradoxical cross-cultural self-concept effect; • attending schools where school-average achievement was high; demonstrating big-fish-little-pond-effects (BFLPE) that generalized over 68 countries, Organisation for Economic Co-operation and Development (OECD)/non-OECD countries, high/low achieving schools, and high/low achieving students; • year-in-school relative to age; unifying different research literatures for associated negative effects for starting school at a younger age and acceleration/skipping grades, and positive effects for starting school at an older age ("academic red shirting") and, paradoxically, even for repeating a grade. Contextual effects matter, resulting in significant and meaningful effects on self-beliefs, not only at the student (year in school) and local school level (BFLPE), but remarkably even at the macro-contextual country-level. Finally, we juxtapose cross-cultural generalizability based on Programme for International Student Assessment (PISA) data used here with generalizability based on meta-analyses, arguing that although the two approaches are similar in many ways, the generalizability shown here is stronger in terms of support for the universality of the frame-of-reference effects."*